Computational Chemistry Comparison and Benchmark DataBase Release 22 (May 2022) Standard Reference Database 101 National Institute of Standards and Technology

Experimental data for N (Nitrogen atom)

22 02 02 11 45

Other names
N; Nitrogen atom; Nitrogen radical; nitrogen;

INChI	INChIKey	SMILES	IUPAC name
InChI=1S/N	QJGQUHMNIGDVPM-UHFFFAOYSA-N	[N]	

State	Conformation
^4S	KH

Enthalpy of formation (Hfg), Entropy, Integrated heat capacity (0 K to 298.15 K) (HH), Heat Capacity (Cp)

Property	Value	Uncertainty	units	Reference	Comment
Hfg(298.15K) $\Delta_f H°$	472.68	0.40	kJ mol^{-1}	CODATA	
Hfg(0K) $\Delta_f H°$	470.82	0.40	kJ mol^{-1}	CODATA	
Entropy (298.15K) $S°$	153.30	0.00	J K^{-1} mol^{-1}	CODATA	
Integrated Heat Capacity (0 to 298.15K) $H°-H°(T)$	6.20	0.00	kJ mol^{-1}	CODATA	
Heat Capacity (298.15K) $C_p°$	20.79		J K^{-1} mol^{-1}	Gurvich	

Information can also be found for this species in the NIST Chemistry Webbook

Electronic energy levels (cm^{-1})

Energy (cm^{-1})	Degeneracy	reference	description
0	4	1949MOO	^4S
19223	6	1949MOO	^2D
19231	4	1949MOO	

Ionization Energies (eV)

Ionization Energy	I.E. unc.	vertical I.E.	v.I.E. unc.	reference
14.534				webbook

Proton Affinity (kJ mol^{-1})

Proton Affinity	unc.	Product	reference	comment
342.2		NH^+	webbook	

Dipole, Quadrupole and Polarizability

Electric dipole moment

State	Config	State description	Conf description	Exp. min.	Dipole (Debye) x	y	z	total	Reference	comment	Point Group	Components dipole	quadrupole
1	1	^4S	K_h	True	0.000	0.000	0.000	0.000	_		K_h	0	0
2	1	^2D	K_h	True							K_h	0	0

Experimental dipole measurement abbreviations: MW microwave; DT Dielectric with Temperature variation; DR Indirect (usually an upper limit); MB Molecular beam

Calculated electric dipole moments for N (Nitrogen atom).

Electric quadrupole moment

State	Config	State description	Conf description	Exp. min.	Quadrupole (D Å)			Reference	comment	Point Group	Components	
					xx	yy	zz				dipole	quadrupole
1	1	^4S	K_h	True						K_h	0	0
2	1	^2D	K_h	True						K_h	0	0

Calculated electric quadrupole moments for N (Nitrogen atom).

Electric dipole polarizability (Å^3)

alpha	unc.	Reference
1.100	0.015	1977Mil/Bed:1

Calculated electric dipole polarizability for N (Nitrogen atom).

References

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squib	reference	DOI
1949MOO	CE Moore "Atomic Energy Levels" NBS Circular 467 (See online version: https://www.nist.gov/pml/atomic-spectra-database)	10.18434/T4W30F
1977Mil/Bed:1	TM Miller, B Bederson, "" Adv. At. Mol. Phys. 13, 1, 1977	10.1016/S0065-2199(08)60054-8
CODATA	Cox, J.D.; Wagman, D.D.; Medvedev, V.A.CODATA Key Values for Thermodynamics. Hemisphere, New York, 1989	10.1002/bbpc.19900940121
Gurvich	Gurvich, L.V.; Veyts, I. V.; Alcock, C. B., Thermodynamic Properties of Individual Substances, Fouth Edition, Hemisphere Pub. Co., New York, 1989	
webbook	NIST Chemistry Webbook (http://webbook.nist.gov/chemistry)	10.18434/T4D303

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SiH5+ (Silane, protonated)	N- (Nitrogen atom anion)